

Neural operator-enabled forward and inverse modeling of laser-induced surface acoustic waves and applications in nondestructive evaluation

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ABSTRACT

Laser-induced surface acoustic wave (SAW)-driven nondestructive evaluation offers high-resolution, non-contact characterization of subsurface microstructures. However, its practical application is often limited by the high computational costs associated with traditional numerical simulation methods. Recently, machine learning has emerged as an attractive alternative to accelerate these simulations. In this paper, we develop a neural operator-enabled framework for both forward and inverse modeling of laser-induced SAW propagation. A general dataset with randomly generated subsurface structures is used to evaluate and quantify the model's performance in both wave propagation and subsurface inversion problems. Three potential applications are then investigated: subsurface crack localization, multilayer structure characterization and polycrystalline grain imaging. The results demonstrate that the neural operator-enabled model achieves satisfactory accuracy even in the presence of noise and source waveform variations, underscoring its potential as an efficient and accurate surrogate model for practical nondestructive evaluation using laser-induced SAWs.

1. Introduction

Ultrasonic guided waves have attracted significant interest across a broad spectrum of fields, ranging from nondestructive evaluation and structural health monitoring of industrial systems (Tian et al., 2019; Capineri and Bulletti, 2021; Yang et al., 2023) to fundamental research on various waveguide configurations (Feng et al., 2024; Hess et al., 2014; M. Liu et al., 2024; Wu et al., 2023) and advanced theoretical modeling methods (Nicoli et al., 2022; X. Liu et al., 2024; Hua et al., 2024; Liu et al., 2025). Acoustic waves can be guided by different types of boundaries, which ultimately determines their application scenarios. Surface acoustic waves (SAWs) (Rayleigh, 1885) are guided waves that use the surface as a waveguide, confining elastic energy to the surface with a penetration depth that depends on the wavelength. This unique characteristic makes broadband SAWs particularly effective for characterizing subsurface features of materials or structures. While SAWs are typically excited by piezoelectric transducers, advances in laser technology have enabled the rapid development of laser-induced SAWs, which offer significantly higher spatial resolution and enable non-contact measurement, leading to widespread applications (Hess et al., 2014; Liu et al., 2022; L. Ding et al., 2023; Huang et al., 2023; Baher et al., 2024; Cheng et al., 2024; Jia et al., 2024; Zarei and Pilla,

2024).

Research on laser-induced SAW-driven nondestructive evaluation is mainly based on analytical modal analysis (Liu et al., 2022; Huang et al., 2023; Jia et al., 2024; Baher et al., 2024) or numerical dynamic analysis (Cheng et al., 2024; L. Ding et al., 2023; Zarei and Pilla, 2024) of waves. Modal analysis, or more specifically dispersion analysis, is valued for its tractability of analysis and ease of implementation, but is often limited in the types of features that can be detected compared to numerical dynamics methods. Traditional numerical methods, such as the finite difference (Liu et al., 2021a), finite element (Cheng et al., 2024; L. Ding et al., 2023; Zarei and Pilla, 2024) and spectral element (Fiborek et al., 2022) methods, simulate wave propagation in complex media by discretizing the elastic wave equation in time and space, approximating its derivatives using finite-difference formulas or basis-function expansions and associated parameters, and performing repeated forward model runs. Although these methods can achieve high solution accuracy, they usually incur significant computational cost, especially in inverse problems, which greatly limits their application in real-world nondestructive evaluation.

Thanks to significant advances in artificial intelligence, particularly in machine learning (ML), many researchers have begun exploring ML-enabled approaches to address wave propagation problems. A

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prominent example is the physics-informed neural network (PINN), which uses a deep neural network to approximate the solution of the equation and leverages auto-differentiation to compute exact derivatives for optimizing a physics-informed loss function. PINNs have achieved remarkable success in solving various wave propagation problems, such as the acoustic and elastic wave equations in isotropic and anisotropic media (Song et al., 2021; Rasht-Behesht et al., 2022; Y. Ding et al., 2023; Ren et al., 2024; Thakur et al., 2024; Guo et al., 2025), the Helmholtz equation (Zhang et al., 2025) and the shallow water equation (Hou et al., 2024). However, PINNs are typically limited to solving a specific problem instance and struggle to generalize across different scenarios, such as changes in source characteristics or velocity structures, without re-optimization. This constraint hinders their effectiveness as surrogate models for practical applications.

Neural operators have emerged as a paradigm for learning solution operators rather than solving individual physical systems. These neural networks treat both inputs and outputs as functions and operate directly on them. For wavefield prediction problems, neural operators learn the solution operators of a family of partial differential equations governing wave propagation. Specifically, the input can encompass various material properties, source parameters, and initial and boundary conditions, while the output represents the corresponding wavefield solutions, capturing both spatial and temporal information. Currently, the Fourier Neural Operator (Li et al., 2021) and its variants, such as the U-shaped Neural Operator (Rahman et al., 2023), the Group-equivariant Fourier Neural Operator (Helwig et al., 2023) and the Factorized Fourier Neural Operator (Tran et al., 2023), have been increasingly applied to such problems in geophysics (Lehmann et al., 2023, 2024; Yang et al., 2021), gradually demonstrating superior performance.

In this paper, we aim to investigate the potential applications of neural operators in ultrasonic nondestructive evaluation. Specifically, we develop a framework for both forward and inverse modeling of laser-induced SAW propagation using neural operators. The remainder of this paper is organized as follows: Section 2 presents the methodology, including dataset generation, operator-enabled modeling and training strategies; Section 3 evaluates the performance of neural operators on a general example, where subsurface features are characterized by a random velocity distribution; in Section 4, we apply the neural operator-enabled approach to tackle challenges encountered in practical laser-induced SAW-driven nondestructive evaluation; conclusions are finally presented in Section 5.

2. Methods

This section aims to design neural operators for forward and inverse modeling of laser-induced SAWs, with a focus on uncovering the intrinsic relationships between SAW waveform evolution and subsurface material properties in a supervised learning setting. The designed operators are intended to serve as highly accurate and efficient surrogates to traditional numerical solvers in applications of ultrasonic testing.

2.1. Laser-induced surface acoustic wave propagation

The generation and propagation of SAWs induced by pulsed laser radiation in an inhomogeneous isotropic medium can be described by solving the governing equation (1) subject to the traction-free boundary condition (2), which are given as follows:

$$\nabla \cdot [\lambda(\nabla \cdot \mathbf{u})\mathbf{I} + \mu(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] + \mathbf{f} = \rho \frac{\partial^2 \mathbf{u}}{\partial t^2}, \quad (1)$$

$$[\lambda(\nabla \cdot \mathbf{u})\mathbf{I} + \mu(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] \cdot \mathbf{n} = \mathbf{0}, \quad (2)$$

where $\mathbf{u}(x, y, z, t)$ is the displacement of the material particle at position (x, y, z) and time t , $\rho(x, y, z)$ is the density, $\lambda(x, y, z)$ and $\mu(x, y, z)$ are the Lamé parameters, $\mathbf{f}(x, y, z, t)$ represents the equivalent mechanical load

at the surface area of laser illumination, and $\mathbf{n}(x, y, z)$ is the outward normal vector on the boundary surface. The nabla symbol ∇ represents the vector differential operator, and the superscript T indicates the transpose for vectors.

When a laser pulse irradiates a solid surface, the surface region absorbs the electromagnetic energy from the laser, resulting in a localized temperature rise. This rapid increase in surface temperature induces thermal expansion, generating ultrasonic waves. To enhance the signal-to-noise ratio (SNR) while maintaining a low energy density to avoid ablation, the laser is often focused into a line source in detection experiments (Jia et al., 2024; Liu et al., 2022). According to Achenbach et al. (Achenbach, 2003; Arias and Achenbach, 2003), a thermoelastic thin line source on the surface of an isotropic material can be modeled as a single dipole parallel to the surface, with its direction perpendicular to the line source. The strength of the dipole is determined by the heat input and material properties. The spatial variation of the dipole can be neglected, and its temporal distribution is often approximated as $(8t^3/\tau^4)e^{-2t^2/\tau^2}$, where τ represents the rise-time parameter of the laser pulse. In this case, if the material along the propagation path of the acoustic field generated by the line source exhibits negligible lateral variation, Eq. (1) can be reduced to a two-dimensional (2D) form by disregarding the y -direction.

To generate the datasets for neural operator training, the propagation of laser-induced SAWs governed by the 2D form of Eq. (1) is solved numerically using the finite difference (FD) method-based software, Deepwave (Richardson, 2023) (see Fig. 1(a)). The half-space propagation domain is modeled as a $1.28 \text{ mm} \times 2.56 \text{ mm}$ rectangle, with a total simulation time of $1.28 \text{ } \mu\text{s}$. The W formulation with the adjusted FD approximations (W-AFDA) scheme (Kristek et al., 2002) is applied on the top side for modeling a free surface, and absorbing boundary conditions employing the convolutional perfectly matched layer (C-PML) method (Martin et al., 2008) are used on the other three sides, to effectively simulate SAW propagation and minimize reflections. The grid spacings Δx and Δz are set to $20 \text{ } \mu\text{m}$, while the time step Δt is set to 5 ns . A single equivalent laser source is placed at the upper-left corner of the model, with $\tau = 150 \text{ ns}$, corresponding to a dominant frequency of approximately 4 MHz for the generated SAWs, and 128 receivers, matching the number of horizontal grid points, are uniformly distributed along the entire top surface of the model to record the SAW wavefield. To reduce the degrees of freedom while reflecting the properties of real-world samples as accurately as possible, the values of density ρ and Poisson's ratio ν are fixed at 8000 kg/m^3 and 0.3 , respectively, which are typical for metals. The only variable reflecting inhomogeneity is chosen to be the shear wave velocity $V_s(x, z)$ of the material. The wavefield described by the vertical velocity component $V_z(x, z, t)$ is recorded as the corresponding counterpart for training. In this work, a total of four different types of datasets are generated, including a general example for validation (Section 3) and three specific examples for application (Section 4).

2.2. Factorized Fourier Neural Operator

An operator $\mathcal{G} : \mathcal{A} \rightarrow \mathcal{U}$ denotes a mapping between two infinite-dimensional function spaces $\mathcal{A}$ and $\mathcal{U}$. The neural operator $\mathcal{G}_\theta$, as a generalization of standard deep neural networks to the operator setting, is designed to learn such mappings using a finite collection of input-output observations. It is usually formulated as the following iterative architecture (Li et al., 2020, 2021):

$$u = \mathcal{G}_\theta(a) = (\mathcal{Q} \circ \mathcal{L}^{(L)} \circ \dots \circ \mathcal{L}^{(1)} \circ \mathcal{P})(a), \quad (3)$$

where $a \in \mathcal{A}$ and $u \in \mathcal{U}$ represent the input and the solution, respectively, θ denotes the parameters of the neural operator, the symbol $\circ$ represents the function composition, $\mathcal{P}$ is the uplift sub-network mapping the input a to the initial latent representation $\mathbf{z}^{(0)}$, $\mathcal{L}^{(\ell)}$ is the ℓ -th nonlinear operator layer, $\mathcal{Q}$ is the projection sub-network mapping the

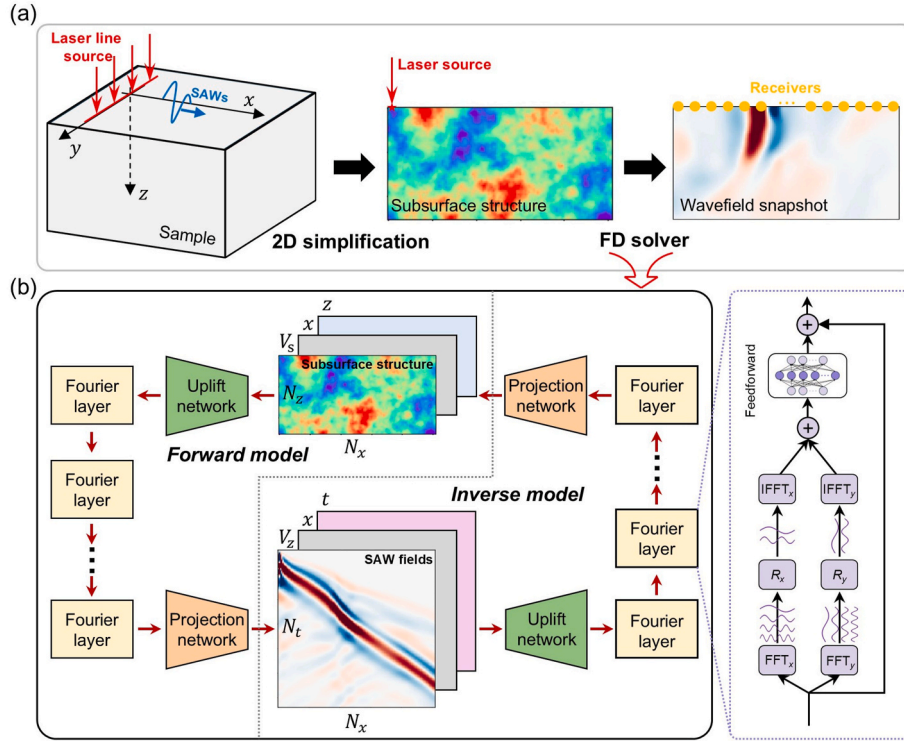


Fig. 1. Schematic of (a) laser-induced SAW dataset generation and (b) neural operator architecture for forward and inverse modeling.

final latent representation $\mathbf{z}^{(L)}$ to the output u , and L is the total number of operator layers.

The type of the operator layer $\mathcal{L}^{(\ell)}$ is not unique, and one specific form that has shown excellent performance in modeling elastic wave propagation (Lehmann et al., 2024) is the factorized Fourier layer (Tran et al., 2023):

$$\mathcal{L}^{(\ell)}(\mathbf{z}^{(\ell-1)}) = \mathbf{z}^{(\ell-1)} + \sigma \left[\mathbf{W}_2^{(\ell)} \sigma \left(\mathbf{W}_1^{(\ell)} \mathcal{K}^{(\ell)}(\mathbf{z}^{(\ell-1)}) + \mathbf{b}_1^{(\ell)} \right) + \mathbf{b}_2^{(\ell)} \right], \quad (4)$$

where σ is a non-linear activation function, $\mathbf{W}_1^{(\ell)}$ and $\mathbf{W}_2^{(\ell)}$ are pointwise linear operators parameterized as matrices, $\mathbf{b}_1^{(\ell)}$ and $\mathbf{b}_2^{(\ell)}$ are bias vectors, and $\mathcal{K}^{(\ell)}$ is the kernel integral operator that factorizes the computation by performing one-dimensional Fast Fourier Transform (FFT) and inverse FFT (IFFT) along each dimension $d \in D$, followed by summation:

$$\mathcal{K}^{(\ell)}(\mathbf{z}^{(\ell-1)}) = \sum_{d \in D} [\text{IFFT}(\mathbf{R}_d^{(\ell)} \cdot \text{FFT}_d(\mathbf{z}^{(\ell-1)}))], \quad (5)$$

where $\mathbf{R}_d^{(\ell)}$ is the Fourier-domain weight matrix.

To serve as an accurate and efficient surrogate model for both forward and inverse modeling of laser-induced SAWs, the neural operator in this work is designed as the Factorized Fourier Neural Operator (F-FNO) (Tran et al., 2023), as described above. The default configuration includes 6 factorized Fourier layers, each with 24 channels, 16 low-order Fourier modes preserved in the spatial dimensions, and 32 low-order Fourier modes preserved in the temporal dimension. The activation function employed is the Rectified Linear Unit (ReLU). The sub-network $\mathcal{P}$ is used to expand the number of input channels to 24 through two dense layers. Conversely, the sub-network $\mathcal{U}$ projects the resulting 24-channel variable $\mathbf{z}^{(L)}$ onto a single-channel physical variable of interest.

2.3. Neural operators for the forward and inverse problems

According to the simulation strategy detailed in Section 2.1, the

subsurface structure model, described by the shear wave velocity V_s , has a size of 64×128 , concatenated with two coordinate grids for x and z . The wavefield, represented by the vertical velocity V_z , has a size of $64 \times 128 \times 256$, concatenated with the same coordinate grids and time t . To reduce computational cost and better align with practical scenarios of forward and inverse problems, only the surface wavefield is retained in this study. In this case, the forward problem is to predict the propagation of SAWs on a given subsurface structure excited by an equivalent laser source. The input a for the forward problem is a variable of size $64 \times 128 \times 3$, comprising three channels: the shear wave velocity V_s , and the x and z coordinates. The output u is a variable of size $64 \times 256 \times 1$, containing only the vertical velocity component V_z as its single channel. In contrast, the inverse problem aims to estimate the subsurface structure from the observed SAW evolution. There are typically two approaches to solve this problem: one is the optimization-based inversion, which iteratively applies the forward operator by updating the structural parameters to minimize the constructed misfit function, similar to the full waveform inversion using traditional numerical solvers (Rao et al., 2020; Aktharuzzaman et al., 2024; Wu et al., 2025); the other is to directly train a new inverse operator, which is simpler and more efficient. Here we adopt the latter approach, where the input a has a format similar to that in the forward problem ($64 \times 256 \times 3$), consisting of V_z , x and t channels, and the output is V_s with a size of $64 \times 128 \times 1$. The overall architecture of forward and inverse modeling is shown in Fig. 1 (b).

It is worth noting that the second dimension of input and output in neural operators undergoes a transformation between space and time. The resulting differences in size are handled by adding or removing zero Fourier coefficients in the high-frequency part of $\mathbf{R}_d^{(\ell)} \cdot \text{FFT}_d(\mathbf{z}^{(\ell-1)})$ (Lehmann et al., 2024). Accordingly, the number of Fourier modes preserved in the factorized Fourier layers is adjusted: 32 low-order Fourier modes are preserved in the five layers adjacent to the time-dependent variable, while 16 low-order Fourier modes are preserved in the remaining layer adjacent to the space-dependent variable. This results in the trained neural operator being capable of efficiently

mapping data between the depth and time dimensions.

2.4. Training approach

After applying appropriate feature scaling to the simulated dataset, the neural operator-enabled model is trained in a supervised learning setting. The dataset is divided into three subsets: 80 % for training, which updates the neural operator parameters; 10 % for validation, which monitors performance and mitigates overfitting; and the remaining 10 % for testing, which evaluates the generalization capability of the model. The training process employs the Adam optimizer (Kingma and Ba, 2017) with hyperparameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$ and the initial learning rate of 0.001, augmented with dynamic learning rate scheduling, where the learning rate is halved after 10 epochs if the validation loss plateaus. Additionally, convergence-driven early termination is applied, with an early stopping patience of 60 epochs and a tolerance threshold of 0.0001. The relative L_1 -norm error serves as the default loss function. The batch size is set to 25, and the maximum number of training epochs is set to 500.

3. Validations

To validate the effectiveness and evaluate the performance of the forward and inverse neural operators for SAW field prediction and subsurface structure imaging, we generate a dataset comprising 2000

samples, where the heterogeneous V_S models, characterized by a mean velocity of 3000 m/s and a standard deviation of 600 m/s, are constructed using the Gaussian random field method with the Matern kernel (Rosofsky et al., 2023), with a smoothness parameter of $\nu = 1$ and a correlation length of $l = 0.256$ mm.

Fig. 2(b) and (a) present a representative subsurface structure and its corresponding SAW field from the test subset, respectively. The trained forward and inverse neural operators are applied to the example, yielding the results shown in Fig. 2(c) and (d). The SAW waveforms and V_S profiles collected at three typical locations are directly compared with true values in Fig. 2(g) and (h). Additionally, Fig. 2(e) and (f) show the errors between the true and predicted values for the forward and inverse problems. The results demonstrate that the forward operator accurately predicts the generation and evolution of SAWs under pulsed laser excitation, as well as the separation of surface skimming longitudinal waves from SAWs, with precise propagation velocity. The inverse operator successfully reconstructs the large-scale subsurface structure variations but does not accurately capture some high frequency details. The subsurface structures closer to the surface are better predicted compared to the deeper ones, which is consistent with the penetration depth of the generated SAWs. For reference, Fig. 2(i) presents complete wavefield snapshots at some representative time points to illustrate the interaction between SAWs and the subsurface structure.

To quantify the accuracy of the forward operator, three quantitative metrics are employed: the time-of-flight (ToF) error for quantifying

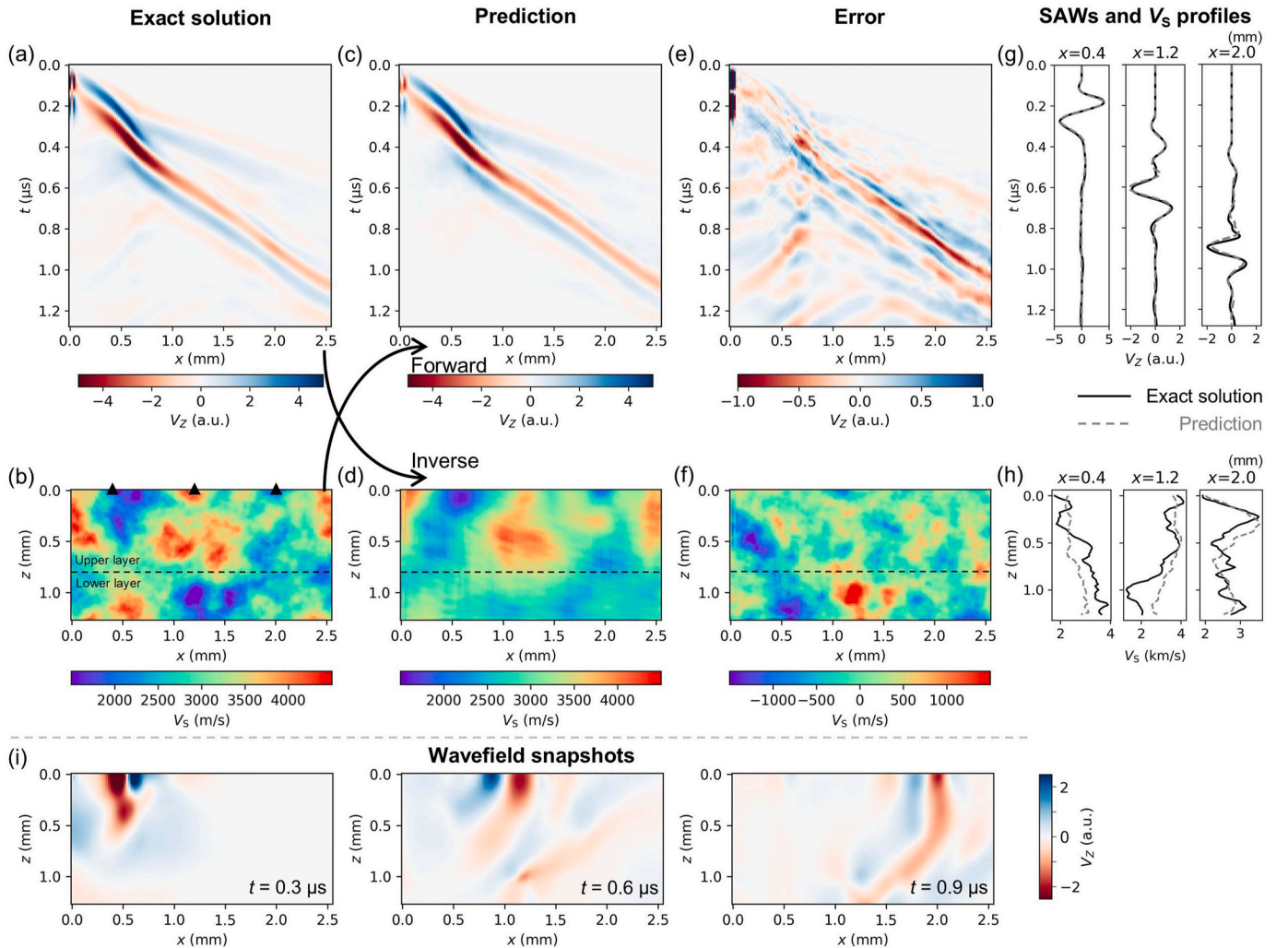


Fig. 2. Results for a test subset of the random V_S dataset. (a) and (b) Ground truth. (c) and (d) Predictions from the forward and inverse operators. (e) and (f) Errors. (g) and (h) Comparison of true and predicted values at three typical positions. (i) Complete wavefield snapshots at some representative time points.

wave velocity accuracy (Eq. (6)), the cross-correlation coefficient for assessing waveform evolution fidelity (Eq. (7)), and relative FFT amplitude of the waveform error for measuring spectral consistency (Eq. (8)):

$$e_{\text{ToF}}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N |\hat{t}_{\max}^{(i)}(\mathbf{x}) - t_{\max}^{(i)}(\mathbf{x})|, \quad (6)$$

$$C_{V_z}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N \frac{\sum_t (\hat{V}_z^{(i)}(\mathbf{x}, t) - \mu_{V_z^{(i)}}(\mathbf{x})) (V_z^{(i)}(\mathbf{x}, t) - \mu_{V_z^{(i)}}(\mathbf{x}))}{\sqrt{\sum_t (\hat{V}_z^{(i)}(\mathbf{x}, t) - \mu_{V_z^{(i)}}(\mathbf{x}))^2 \sum_t (V_z^{(i)}(\mathbf{x}, t) - \mu_{V_z^{(i)}}(\mathbf{x}))^2}}, \quad (7)$$

$$e_{\text{FFT}}(f) = \frac{\sum_{i=1}^N \sum_x |\text{FFT}(\hat{V}_z^{(i)}(\mathbf{x}, t) - V_z^{(i)}(\mathbf{x}, t))|}{\sum_{i=1}^N \sum_x |\text{FFT}(V_z^{(i)}(\mathbf{x}, t))|}, \quad (8)$$

where $t_{\max}^{(i)}(\mathbf{x}) = \arg \max_t V_z^{(i)}(\mathbf{x}, t)$ represents the time at which the SAW signal reaches its maximum value at position $\mathbf{x}$ for sample i , $\mu_{V_z^{(i)}}(\mathbf{x}) = \frac{1}{N_t} \sum_t V_z^{(i)}(\mathbf{x}, t)$ ($N_t = 256$) represents the mean amplitude of the SAW signal at position $\mathbf{x}$ for sample i , f is the frequency, and N is the total number of test samples. The quantities for the exact and predicted values share the same symbols, but the predicted values are distinguished by a hat.

The inverse operator is evaluated using the following two metrics: the cross-correlation coefficient to assess the similarity of the V_s profile shape (Eq. (9)), and the absolute error to quantify the accuracy of the V_s values (Eq. (10)). Furthermore, to analyze depth-dependent reconstruction performance, the subsurface structure is divided into upper and lower layers using 0.8 mm as the interface, and the metrics are separately quantified for each layer.

$$C_{V_s}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N \frac{\sum_z (\hat{V}_s^{(i)}(\mathbf{x}, z) - \mu_{V_s^{(i)}}(\mathbf{x})) (V_s^{(i)}(\mathbf{x}, z) - \mu_{V_s^{(i)}}(\mathbf{x}))}{\sqrt{\sum_z (\hat{V}_s^{(i)}(\mathbf{x}, z) - \mu_{V_s^{(i)}}(\mathbf{x}))^2 \sum_z (V_s^{(i)}(\mathbf{x}, z) - \mu_{V_s^{(i)}}(\mathbf{x}))^2}}, \quad (9)$$

$$e_{V_s}(\mathbf{x}) = \frac{1}{NN_z} \sum_{i=1}^N \sum_z |\hat{V}_s^{(i)}(\mathbf{x}, z) - V_s^{(i)}(\mathbf{x}, z)|, \quad (10)$$

where $\mu_{V_s^{(i)}}(\mathbf{x}) = \frac{1}{N_z} \sum_z V_s^{(i)}(\mathbf{x}, z)$ represents the meanvalue of V_s profile at position $\mathbf{x}$ for sample i , and $N_z = 40$ for upper layer and $N_z = 24$ for lower layer.

Fig. 3 presents the average results across all 200 test samples for the 6-layer, 24-channel configuration. It is observed in Fig. 3(a) and (b) that both the ToF error and cross-correlation coefficient degrade as the propagation distance increases, with average values of $1.575\Delta t$ and 0.977, respectively. This deterioration is attributed to the increasing distortion, spreading and attenuation of the acquired SAW waveform due to scattering caused by the heterogeneous subsurface structure. The relative waveform error in the frequency domain, shown in Fig. 3(c), demonstrates that the errors become more pronounced as the frequency increases, despite the fundamental frequency being 4 MHz. The x-axis range in Fig. 3(c) corresponds to the effective bandwidth, defined by the 1/e threshold of the peak amplitude. The results in Fig. 2(d) and Fig. 3(c) reflect the well-known spectral bias phenomenon, where neural networks tend to first learn low-frequency components corresponding to large-scale features (Rahaman et al., 2019). Layer-wise evaluation results of the inverse operator in Fig. 3(d) and (e) show better accuracy in the upper layer, with a mean correlation coefficient of 0.617 and velocity error of 265.8 m/s, compared to the lower layer (0.214 and 417.8

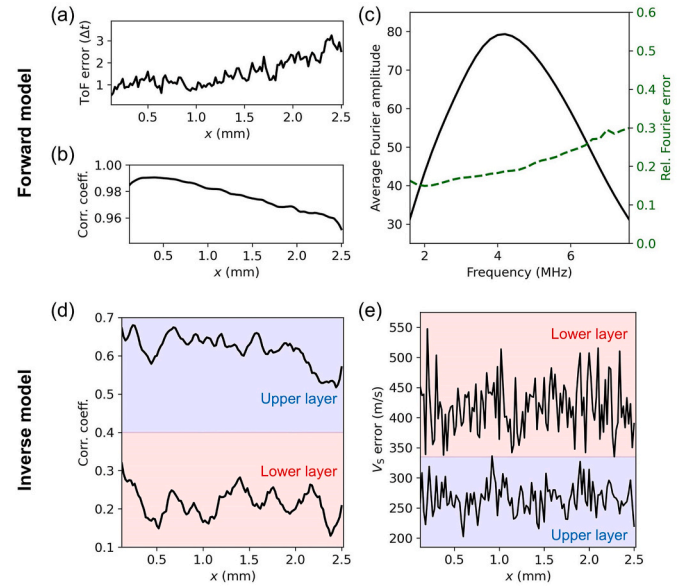


Fig. 3. Quantification of the accuracy of the neural operators on the test subset. (a) ToF error, (b) cross-correlation coefficient, and (c) relative FFT amplitude of the waveform error for the forward operator. (d) Cross-correlation coefficient and (e) velocity error for the inverse operator.

m/s, respectively). The upper-layer correlation degrades with propagation distance due to the increasing complexity in explaining the far-field SAW waveform. In contrast, the other three quantitative results in the inversion do not exhibit clear correlations with propagation distance. Overall, despite a decline in the performance of both the forward and inverse operators with increasing propagation distance, the results remain within an acceptable range.

As with conventional neural network training, hyperparameters influence the performance of neural operators, with the number of channels and layers being particularly impactful on the prediction of elastic wave propagation (Lehmann et al., 2024). We systematically evaluate 10 pairs of forward and inverse operators with different numbers of channels and layers using the metrics defined in Eqs. (6)–(10) (e.g., ToF), and quantitatively compare them with the default configuration from Fig. 3, as summarized in Fig. 4. The results indicate that waveform prediction accuracy consistently improves with larger architectures, reaching an optimal balance at 24 layers and 6 channels. In contrast, the inverse operators primarily depend on channel capacity, while variations in layer depth result in only marginal fluctuations. To effectively resolve SAW evolution details and capture more features in the V_s profile, a high-channel configuration (≥ 24 channels) is necessary. Finally, the performance of the default forward and inverse configurations (24 layers and 6 channels) is evaluated across the training, validation and test datasets using the ToF error and the cross-correlation coefficient as metrics. As shown in Fig. 5, the results for the three datasets are comparable, indicating that the model in this study is not subject to overfitting.

4. Applications

This section explores the practical potential of neural operators in laser-induced SAW-driven nondestructive evaluation through three representative case studies: subsurface crack localization with position and orientation estimation (Section 4.1), gradient characterization of multilayer structures (Section 4.2), and grain structure imaging in polycrystalline materials (Section 4.3). Each case systematically addresses both the forward and inverse problems of wave propagation while examining key practical factors that may affect the neural operator performance, including measurement noise, pulse source waveform

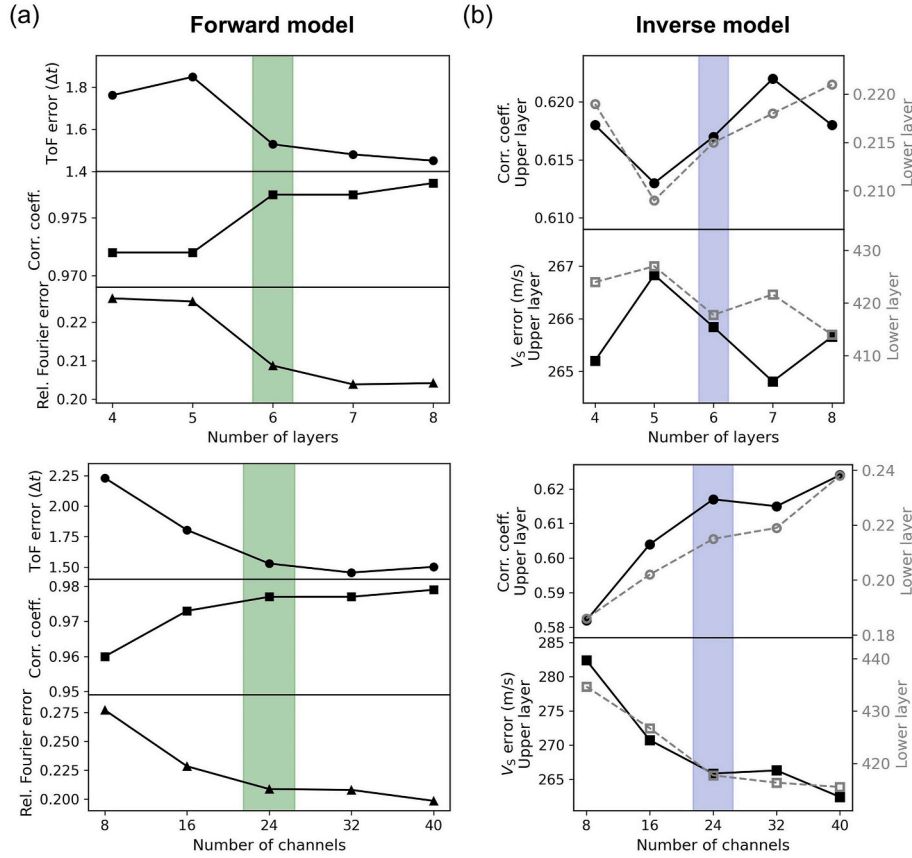


Fig. 4. Accuracy of the (a) forward and (b) inverse operators versus the number of layers and channels. The shaded areas correspond to the default configuration.

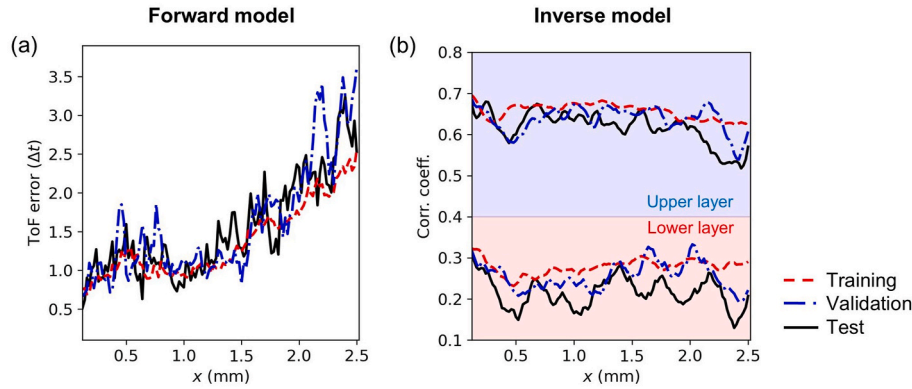


Fig. 5. Evaluation of the default (a) forward and (b) inverse models on training, validation and test datasets.

variations caused by material heterogeneity or laser ablation.

4.1. Subsurface cracks

The first case study examines a homogeneous medium containing a rectangular subsurface crack, which is a representative application of laser-induced SAWs (Hess et al., 2014; Qin et al., 2023; Mao et al., 2024; Cheng et al., 2025). Since the surface typically experiences the highest stress levels, material degradation often initiates there, eventually leading to fatigue or stress corrosion cracking. Real cracks are typically not fully open but have partially contacting surfaces, so they can be simply modeled as the regions of reduced stiffness or velocity (Hess et al., 2014; Shukla et al., 2020). Here, a dataset of 500 samples is generated, where each crack is characterized by a $0.64 \text{ mm} \times 0.08 \text{ mm}$

rectangular region with an associated velocity V_s of 1500 m/s. The crack center positions and orientations are randomly assigned but constrained within the SAW sensitivity region ($z < 0.48 \text{ mm}$).

Fig. 6 presents the results for a representative test subset. In Fig. 6(a), the scattered acoustic field clearly depicts the incident, transmitted and reflected SAWs near the crack. However, due to generally weak reflection signals across the dataset, the trained forward operator fails to capture these crack-related features, indicating that direct inversion of cracks using traditional forward models would likely be unsuccessful. In contrast, the inverse operator automatically and effectively learns the relationship between the weak scattered field and the crack's position and orientation, as demonstrated in Fig. 6(d). Notably, due to abrupt velocity changes at the crack in the V_s data, the loss function used during the inverse operator training is adjusted to a relative L_2 -norm error to

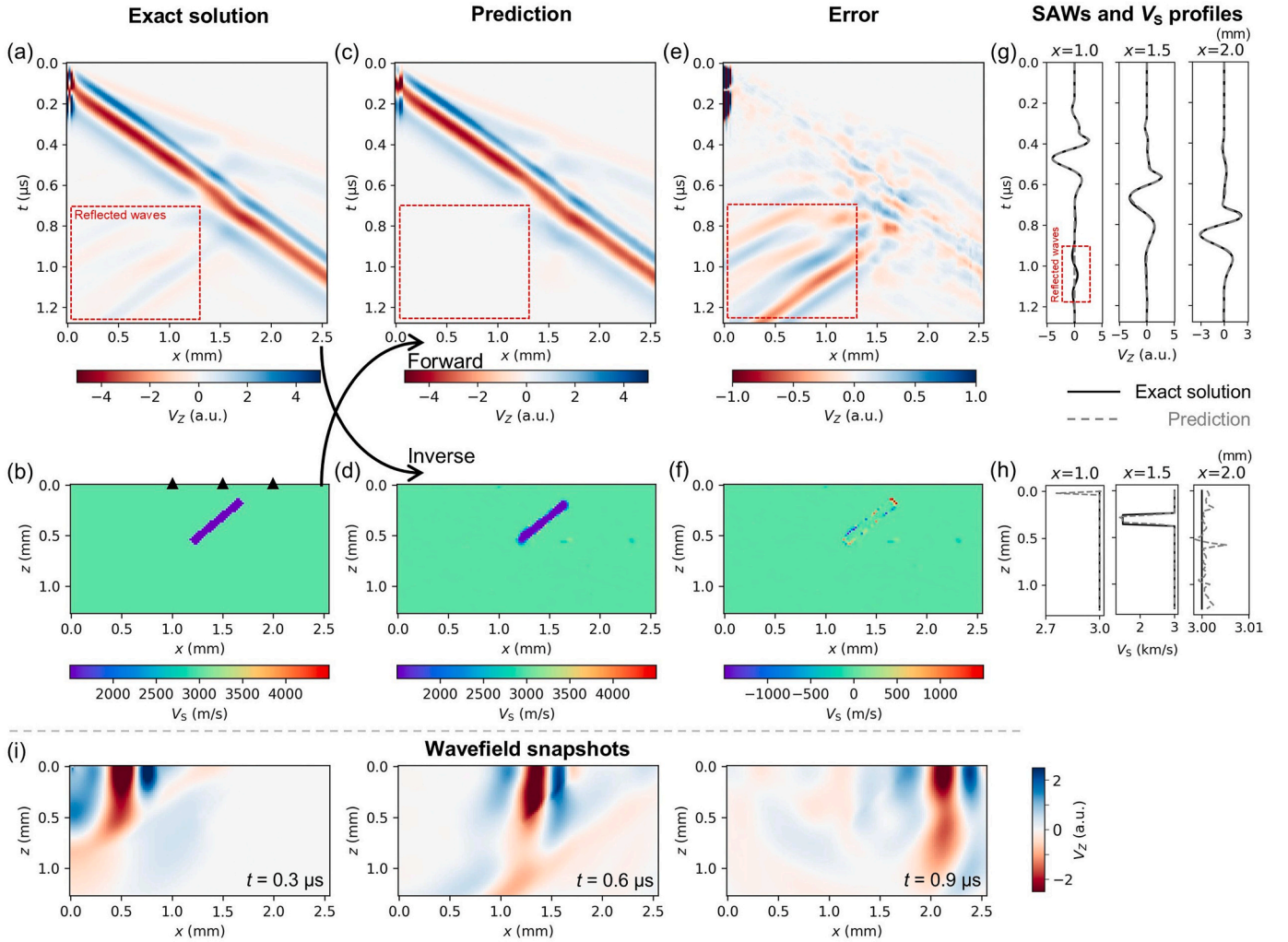


Fig. 6. Results for a test subset of the crack dataset. (a) and (b) Ground truth. (c) and (d) Predictions from the forward and inverse operators. (e) and (f) Errors. (g) and (h) Comparison of true and predicted values at three typical positions. (i) Complete wavefield snapshots at some representative time points.

improve training efficiency. This adjustment is also applied in subsequent cases with similar conditions.

In practice, the quality of the SAW field used for inversion can be affected by instrument limitations or noise from various sources. To test the robustness of the inverse operator, Gaussian white noise with SNR levels of 24, 12 and 6 dB is added to the signal. Fig. 7 compares the inverse operator's subsurface crack imaging predictions under these noise conditions. The results clearly indicate that the quality of crack imaging is directly linked to the quality of the scattered field signal, with cracks becoming indistinguishable only when they are completely overwhelmed by noise. These findings confirm that the trained inverse operator is highly robust to input noise.

All training and testing are conducted under the assumption of a fixed equivalent force source. However, the source signal may be influenced by factors such as material properties, laser intensity, and the extent of thermoelastic and ablation effects, resulting in an unknown source waveform. We have tested that the inverse operator trained above cannot correctly invert the SAW field obtained from other source types, where the change in the source pulse shape is obtained by changing the source directivity. Nevertheless, since the laser duration remains constant while the irradiation width is adjustable, the bandwidth of the resulting pulsed SAW signal can be controlled. Consequently, the inverse operator can be trained by extracting the envelope of the SAW field via the Hilbert transform, thereby establishing its relationship with crack information. Fig. 8 presents the final results,

demonstrating that satisfactory crack imaging is achieved for all three pulse shapes due to their similar wave envelopes, despite some minor artifacts in Fig. 8(f) caused by increased envelope discrepancies.

4.2. Multilayer structures

Next, we focus on another critical application of laser-induced SAWs (Choi et al., 2022; Hess et al., 2014; Huang et al., 2023; Jia et al., 2024; Liu et al., 2022): the gradient characterization of multilayered media, such as coatings or thin films, which is of paramount importance in microelectronics and materials science. To investigate the performance of the neural operator in this context, a dataset of 1000 samples is generated, where the top 0.8 mm of each sample is divided into five layers, each with a uniform thickness of 0.16 mm. The shear wave velocity V_s in each layer is randomly assigned within the range of 1500–4500 m/s to simulate varying material properties.

Fig. 9 presents the results for a representative test subset. The forward operator accurately captures the complex dispersion characteristics of SAW propagation, despite slight increases in waveform complexity leading to minor errors over time. Meanwhile, the inverse operator successfully reconstructs the velocity gradient of the multilayer structure with well-defined interfacial boundaries, regardless of whether the layers are shallow or deep. Since the material properties of the layered medium are independent of the lateral x direction, SAW dispersion accumulates in a stable manner. As a result, even in the

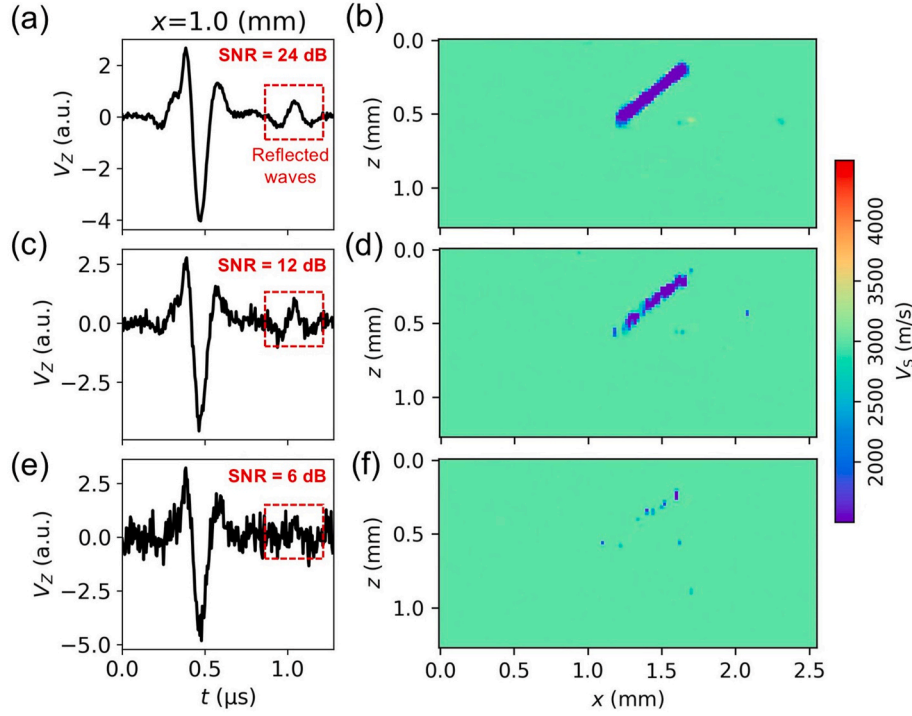


Fig. 7. Impact of noise on crack imaging. (a), (c) and (e) show SAW signals at a representative location with SNR levels of 24, 12 and 6 dB, respectively. (b), (d) and (f) present the corresponding inversion results.

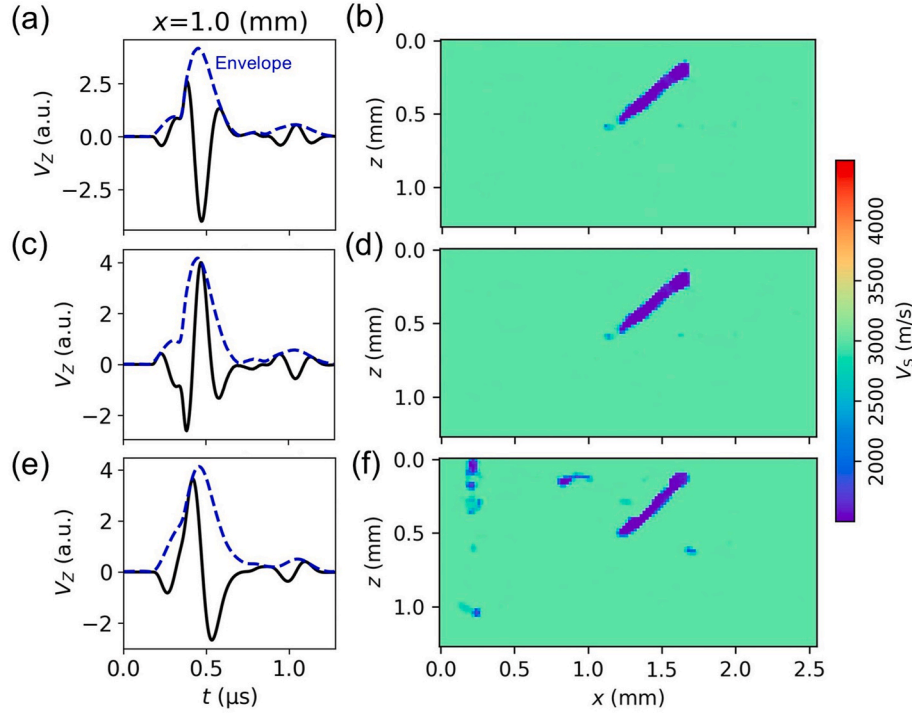


Fig. 8. Inverse operator trained on the SAW envelope field dataset for crack imaging. (a), (c) and (e) show the SAW signals at a representative location for three different source conditions, respectively. (b), (d) and (f) present the corresponding inversion results.

presence of noise in the input of the inverse operator, the velocity gradient of the multilayer structure can still be reliably characterized, as demonstrated in Fig. 10.

The gradient characterization here is achieved through full waveform inversion. However, a more conventional approach involves first extracting the SAW dispersion curve from the time-domain signal and

then performing inversion based on the extracted curve (Liu et al., 2021b; Olafsdottir et al., 2018). This method mitigates challenges associated with variations in the source signal and offers high robustness. To explore this alternative, a simpler one-dimensional (1D) inverse operator is trained, where the input is the SAW dispersion curve, and the output is the V_s profile. These dispersion curves are theoretically

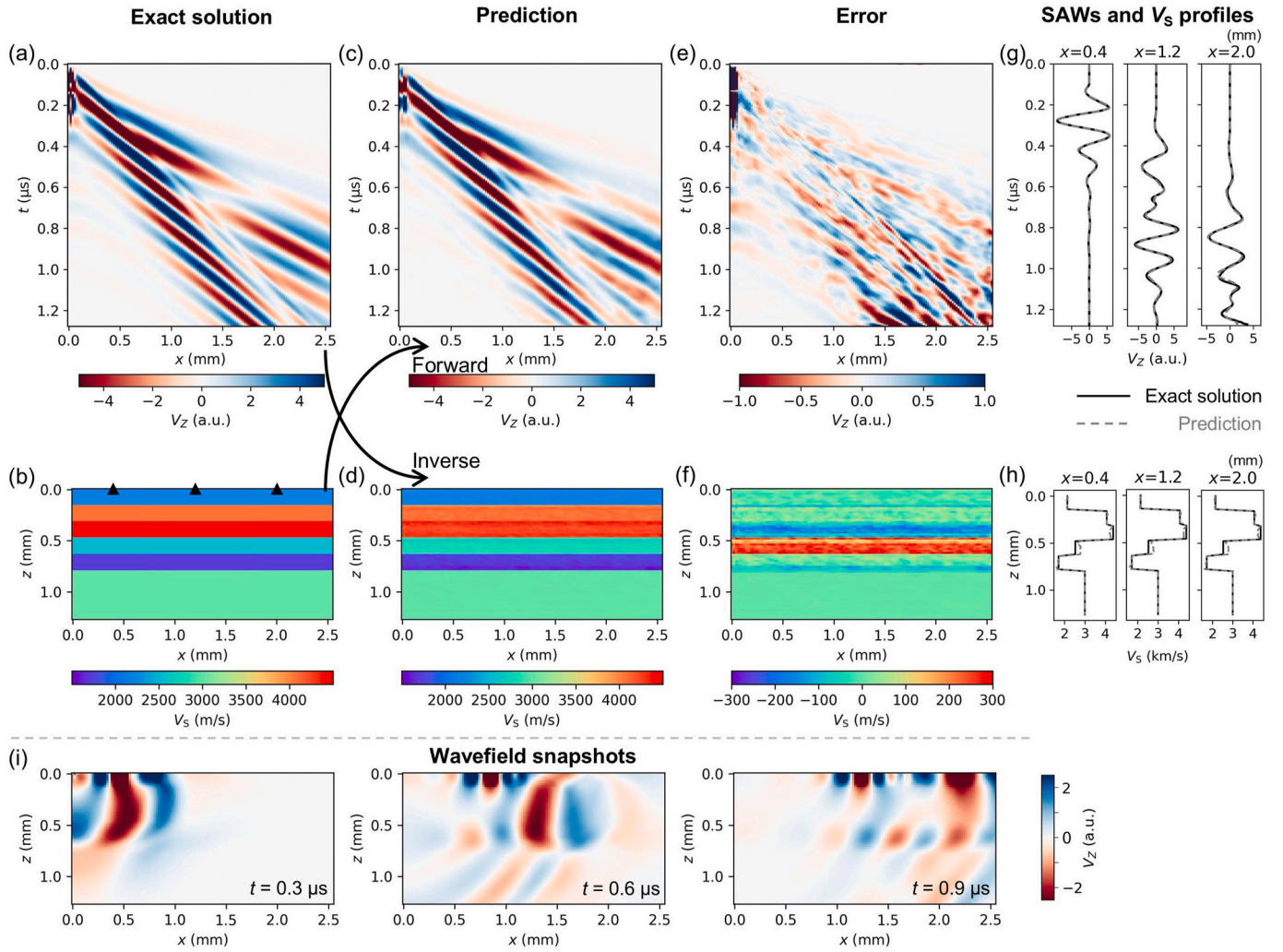


Fig. 9. Results for a test subset of the multilayer structure dataset. (a) and (b) Ground truth. (c) and (d) Predictions from the forward and inverse operators. (e) and (f) Errors. (g) and (h) Comparison of true and predicted values at three typical positions. (i) Complete wavefield snapshots at some representative time points.

calculated using the stiffness matrix method (Rokhlin and Wang, 2002) from the given V_s distribution. Fig. 11 presents the results for three representative test subsets, demonstrating the accuracy of the 1D dispersion curve inversion operator. The prediction errors of this 1D operator are comparable to those of the 2D full waveform inversion operator, but the training process is significantly easier. Therefore, the 1D operator may serve as a more practical alternative for real-world nondestructive testing of multilayer structures.

4.3. Polycrystalline materials

In this final case study, we address a particularly challenging application: grain imaging in polycrystalline materials, capturing both crystallographic orientation and grain morphology (Patel et al., 2017; Dryburgh et al., 2020; Z. Sun et al., 2024; Y.-C. Sun et al., 2024). Since material properties are largely governed by microstructure, accurate microstructural characterization is critical for optimizing materials for specific functions. To explore this, a dataset of 2000 samples is generated by randomly placing seed points followed by Voronoi tessellation, creating approximately 10 grains per sample. Each grain's shear wave velocity V_s is randomly assigned within the range of 1500–4500 m/s, which typically depends on crystallographic orientation. It is important to note that although polycrystalline grain imaging in real materials typically involves micrometer-scale features, the model trained on SAW datasets at different spatial scales remains applicable, owing to the

wavelength-dependent penetration characteristics of SAWs, as long as the SAW wavelength is comparable to the grain size.

Fig. 12 presents representative results in the test subset. The forward operator successfully captures uniform SAW propagation across individual grain regions following the passage of the wave through grain boundaries, although it still exhibits significant prediction errors for weak reflected waves. The inverse operator accurately identifies the primary morphological features of the grains, delineating three distinct regions near the surface: grains A, B, and a combined region comprising C, D and E. Notably, while the grain boundaries in region A and region B are clearly resolved, the boundaries in the third region (containing grains C, D and E) are nearly indiscernible due to minimal impedance contrast between the grains, leading to only weak variations in the SAW evolution. Fig. 12(f) displays the inversion error distribution, showing that discrepancies are primarily concentrated near grain boundaries where velocity discontinuities occur. This is still attributed to the spectral bias of the neural operator, as discussed earlier.

Fig. 13 further illustrates the impact of noise on grain imaging. As noise levels increase, the SAW features and velocity information associated with grain boundaries become increasingly ambiguous, leading to a degradation in image clarity. Nevertheless, the three primary regions can still be vaguely distinguished. Notably, the vertical striping artifacts observed in Fig. 13(d) and (f) originate from the local nature of the inversion process, where the subsurface structure at each lateral position is inferred from the SAW waveform differences recorded by neighboring

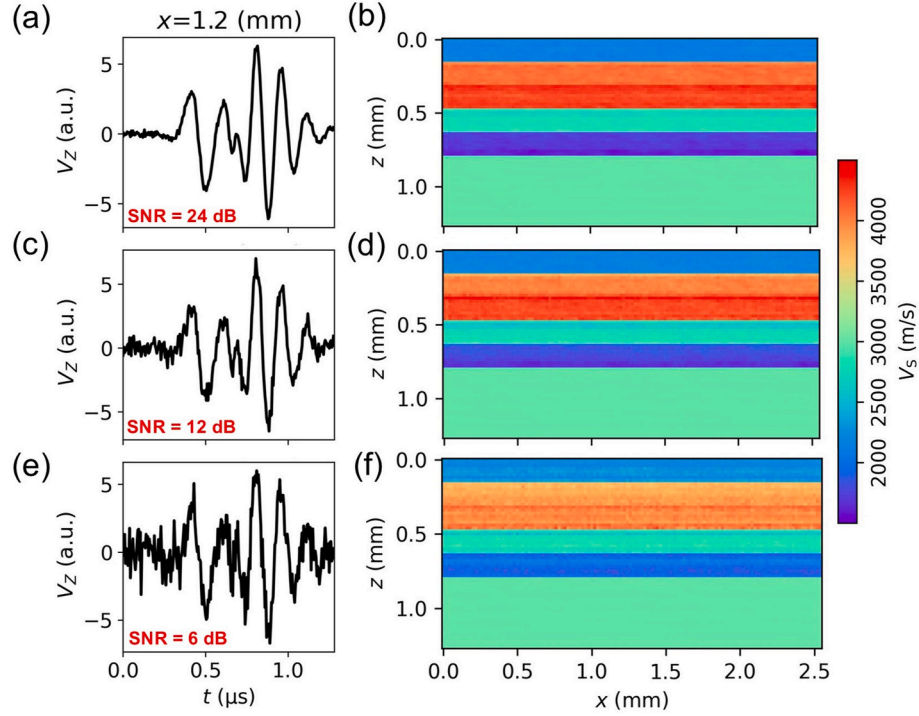


Fig. 10. Impact of noise on the characterization of multilayer structure. (a), (c) and (e) show SAW signals at a representative location with SNR levels of 24, 12 and 6 dB, respectively. (b), (d) and (f) present the corresponding inversion results.

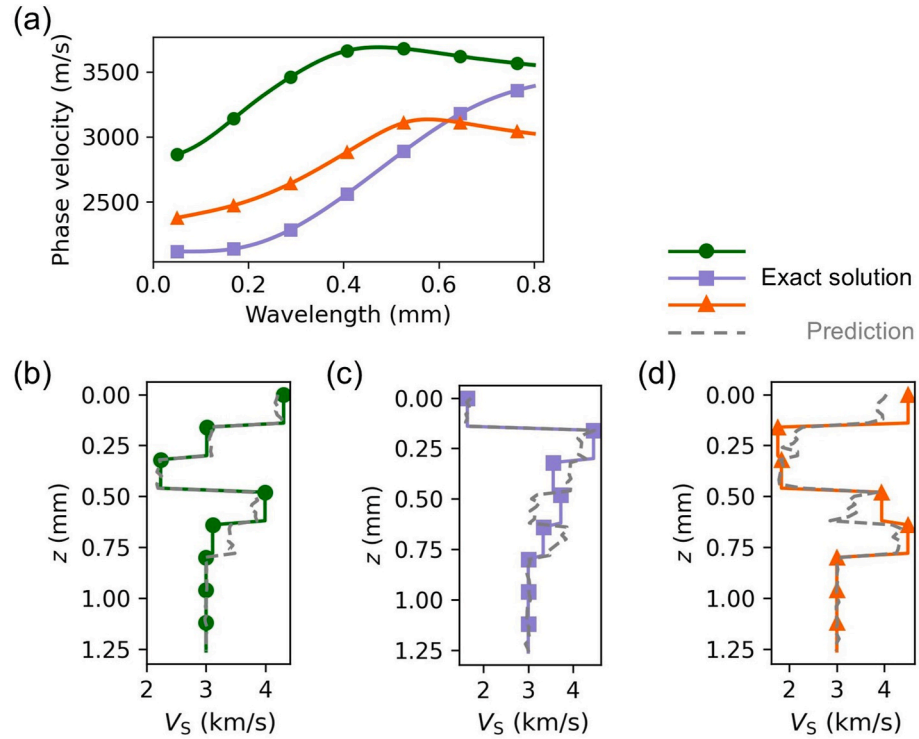


Fig. 11. 1D inverse operator trained on the dispersion curve dataset for multilayer structure characterization. (a) Theoretical dispersion curves. (b), (c) and (d) Predictions from the 1D inverse operator.

receivers. In the presence of noise, these waveform distortions vary across receivers, leading to inconsistencies between adjacent profiles and producing vertically aligned artifacts in the reconstructed images.

Finally, we examine the influence of source waveform variations. Unlike the previous cases, where operator training is modified by pre-

processing the SAW fields, here we adopt a more general approach by generating a new dataset for polycrystalline materials under random-shaped pulse sources by changing the equivalent force source direction. In this setup, the input remains the SAW field, and the output is the V_s profile, without incorporating any source function information.

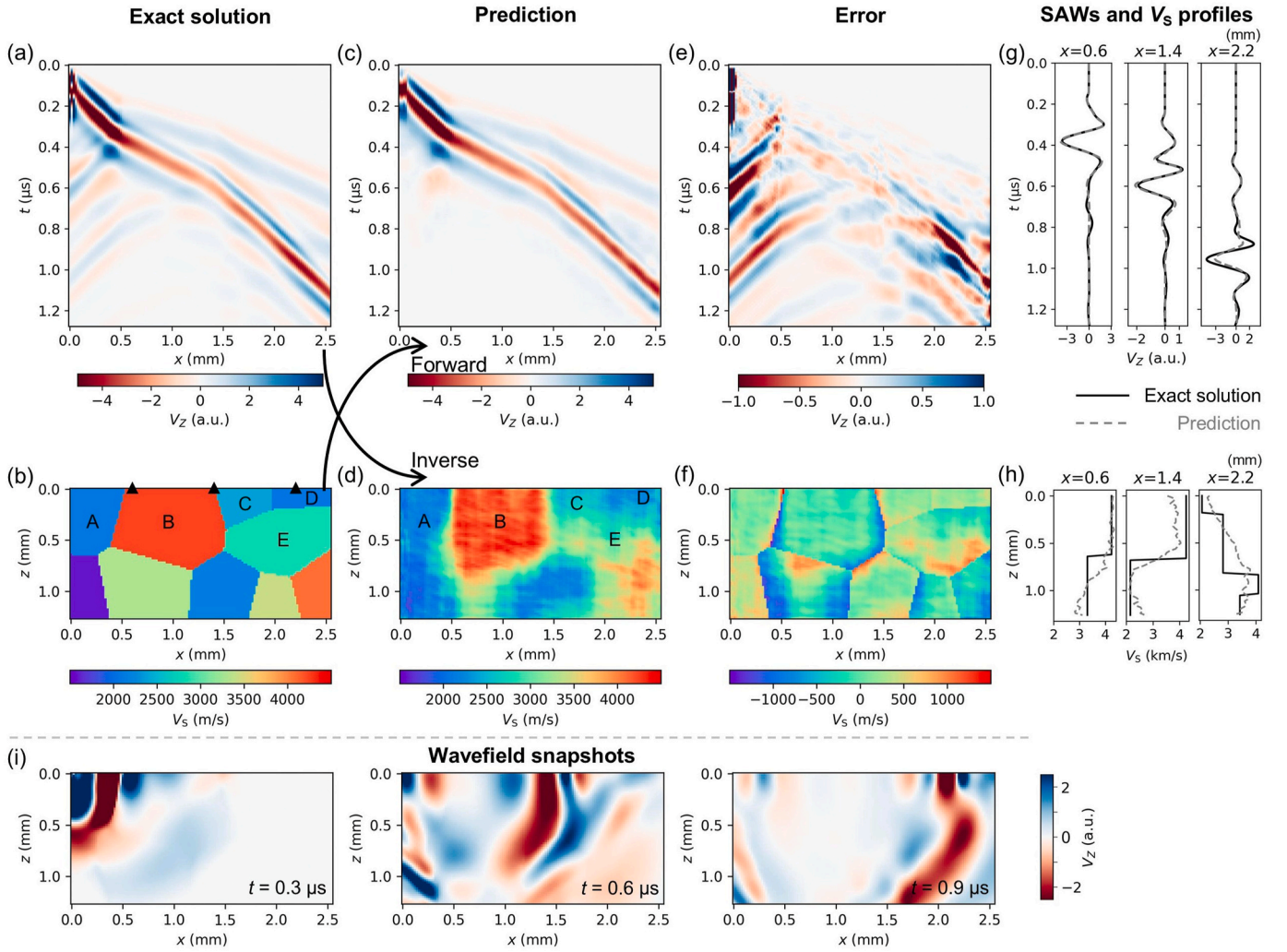


Fig. 12. Results for a test subset of the polycrystalline material dataset. (a) and (b) Ground truth. (c) and (d) Predictions from the forward and inverse operators. (e) and (f) Errors. (g) and (h) Comparison of true and predicted values at three typical positions. (i) Complete wavefield snapshots at some representative time points.

Fig. 14 shows the results of applying the trained inverse operator to the same microstructure under different source conditions. Although imaging of deeper grains exhibits some distortion, near-surface imaging and grain boundary delineation remain clear, indicating that the operator can naturally learn to strip away the source effects and explore the intrinsic grain-related SAW evolution information.

5. Conclusions

This study presents a novel modeling approach enabled by neural operators for the forward and inverse simulation of laser-induced surface acoustic wave (SAW) propagation in nondestructive testing. In our method, both the forward and inverse neural operators are constructed using uplift and projection layers in combination with factorized Fourier layers and trained in a supervised manner. The ground truth wavefield data, represented as a three-dimensional variable with two spatial and one temporal dimension, is obtained from numerical simulations of elastic wave propagation from a fixed equivalent laser source. To maintain computational efficiency and practical relevance, only the essential SAW field information is retained. The performance of the operators is evaluated using a random velocity dataset and some metrics such as the time-of-flight error, demonstrating their good generalization capabilities in this context.

Building on the promising validation results, the neural operator-enabled method is subsequently applied to representative laser-

induced SAW-driven nondestructive evaluation scenarios, aiming to address key challenges in the field. Comprehensive assessments on subsurface crack localization, multilayer structure characterization and polycrystalline grain imaging demonstrate that the forward operator effectively captures the complex scattering and dispersion of SAW propagation, while the inverse operator reliably reconstructs velocity profiles reflecting microstructural features even in the presence of noise and source waveform variations. Although inherent spectral bias issues may exist within the neural operators, their overall performance remains highly satisfactory. Moreover, the dispersion curve-based one-dimensional inverse operator achieves prediction errors comparable to full waveform inversion while significantly reducing training complexity, making it particularly well-suited for multilayer structures. These findings underscore the versatility, robustness and efficiency of neural operators in capturing complex wave dynamics and microstructural characteristics, thereby paving the way for their broader application in practical nondestructive evaluation.

CRediT authorship contribution statement

Zaiwei Liu: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Zhongqing Su:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

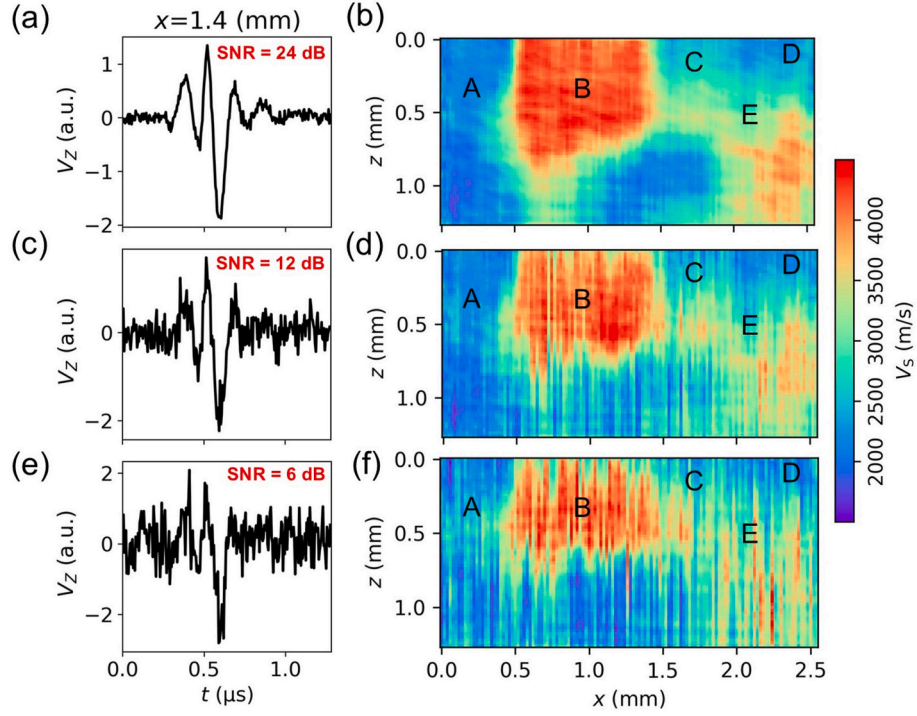


Fig. 13. Impact of noise on the imaging of polycrystalline structure. (a), (c) and (e) show SAW signals at a representative location with SNR levels of 24, 12 and 6 dB, respectively. (b), (d) and (f) present the corresponding inversion results.

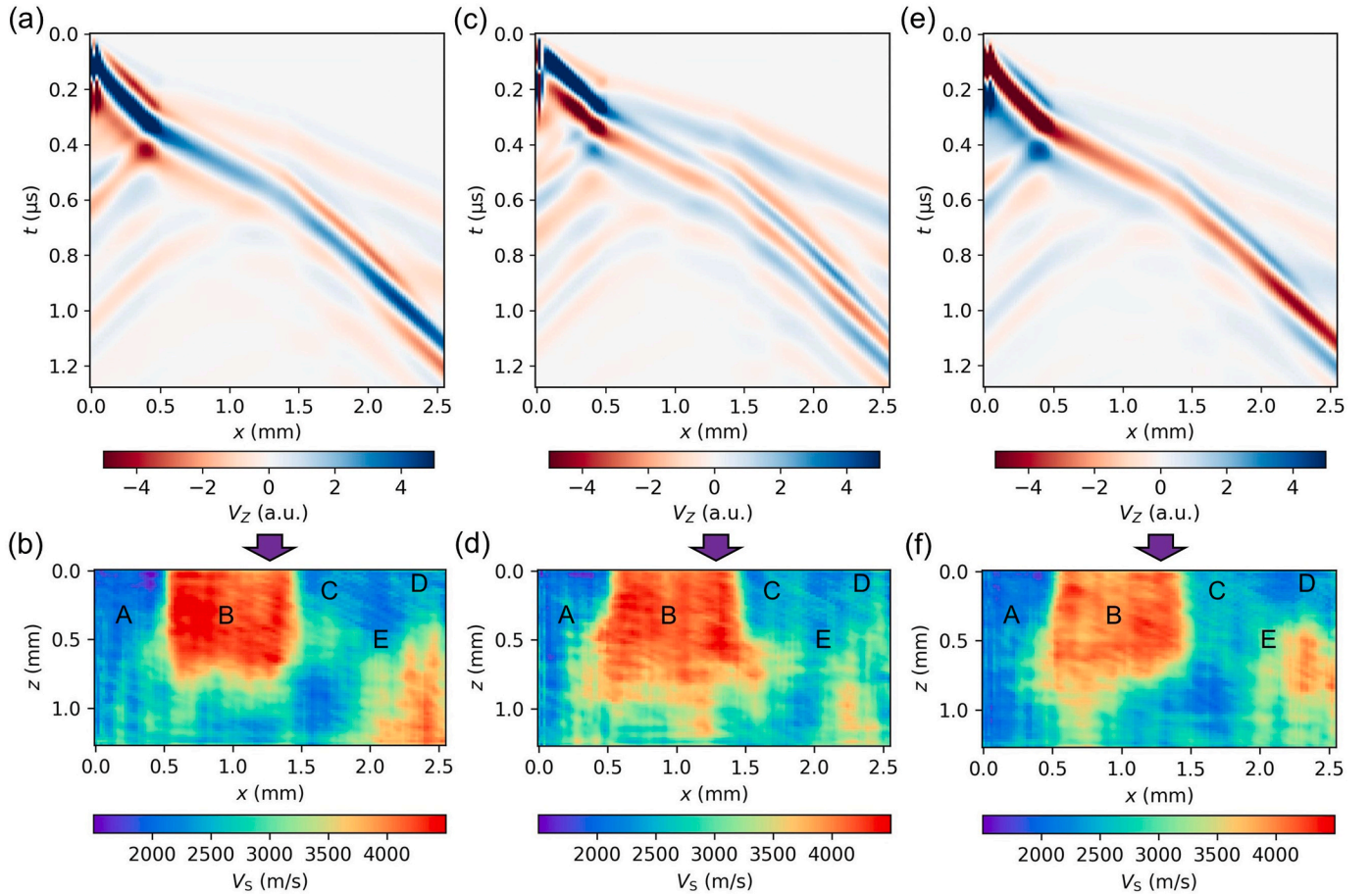


Fig. 14. Inverse operator trained on the dataset of polycrystalline materials under arbitrary pulse sources for grain imaging. (a), (c) and (e) show the SAW fields obtained under three different source waveforms, respectively. (b), (d) and (f) present the corresponding inversion results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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